Pattern-Expert Additive Models: Interpretable Classifiers that Win on Localized High-Order Interactions

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Abstract

Generalized additive models with pairwise interactions (GA2M), such as the Explainable Boosting Machine (EBM), are the accuracy leaders among intrinsically interpretable models for tabular classification, but they cannot represent interactions of order three or higher. Black-box gradient boosting captures such interactions yet requires post-hoc, approximate explanations. We study the *Pattern-Expert Additive Model* (PEAM), which partitions features into groups, fits one calibrated, arbitrarily flexible expert per group, and combines the experts additively in log-odds space. Each prediction decomposes *exactly* into signed per-group reason codes, with no post-hoc approximation; interactions of any order are captured *within* a group, while the model stays additive *across* groups. We show — honestly — that PEAM is not a general-purpose state of the art: on linear, pairwise, saturated, or cross-group-interaction data it is matched or beaten by simpler models. Instead it occupies a precise and previously uncharted point on the accuracy–interpretability frontier. Through a controlled synthetic study that sweeps interaction order, and experiments on four public datasets, we characterize exactly when PEAM wins: when the predictive signal consists of *high-order interactions localized within a known feature grouping*. There, PEAM is the best interpretable model and can beat strong black boxes. On the MADELON benchmark PEAM attains ROC-AUC 0.90 versus 0.80 for EBM and 0.86 for full-feature XGBoost; on HIGGS, whose high-order signal is cross-object (delocalized), PEAM correctly fails — a contrast that isolates locality, not merely interaction order, as the operative condition. We also show that which grouping method succeeds is structure-dependent, and that neither correlation nor univariate mutual information is universally adequate.

1 Introduction

Tabular classifiers face a persistent tension between accuracy and interpretability. At one end, generalized additive models (Hastie & Tibshirani, 1990) and their modern incarnations — in particular GA2M and the Explainable Boosting Machine (EBM) (Lou et al., 2013; Caruana et al., 2015; Nori et al., 2019) — are intrinsically interpretable: every prediction is a sum of univariate shape functions and (for GA2M) pairwise interaction terms, each of which can be plotted and audited. At the other end, gradient-boosted tree ensembles (Chen & Guestrin, 2016; Ke et al., 2017) achieve higher accuracy by modelling interactions of arbitrary order, but they are black boxes whose explanations must be reconstructed post hoc with methods such as SHAP (Lundberg & Lee, 2017) or LIME (Ribeiro et al., 2016), which are approximate and can be unfaithful to the underlying model.

A structural limitation sits at the heart of this trade-off: GA2M, by construction, is additive in main effects and *pairwise* interactions. It *cannot* represent a genuine third- or higher-order interaction. When the label depends on such an interaction, GA2M is misspecified and a black box wins — but then interpretability is lost. This motivates a simple question: *is there an intrinsically interpretable model that captures high-order interactions exactly where they occur, while producing faithful attributions?*

We study one answer, the **Pattern-Expert Additive Model (PEAM)**. PEAM partitions the features into K groups, fits a single calibrated but otherwise *arbitrarily flexible* expert on each group, and combines the experts in log-odds (evidence) space:

$$\text{logit } p(x) \;=\; b_0 \;+\; \sum_{k=1}^{K} \beta_k \left[\text{logit } e_k(x_{G_k}) - \text{logit } \pi \right], \tag{1}$$

where e_k is the group-k expert and π the base rate. Because the combination is additive, every prediction decomposes *exactly* into signed per-group reason codes $r_k = \beta_k[\text{logit } e_k - \text{logit } \pi]$ that sum to the decision log-odds — no post-hoc approximation. Because each expert is unconstrained, interactions of *any* order are captured *within* a group; because the experts are combined additively, the model assumes no interaction *across* groups.

Our central message is deliberately honest and double-edged. PEAM is **not** a general-purpose state of the art. On data that is linear, dominated by pairwise interactions, near-separable, or whose interactions span groups, PEAM is matched or beaten by logistic regression, by EBM, or by a black box. What PEAM offers is a *new, precisely characterizable* operating regime: when the predictive signal is composed of *high-order interactions that are localized within a known grouping*, PEAM's inductive bias is exactly right, and it becomes the best interpretable model — at times beating even black boxes, because its structure additionally performs implicit structured feature selection. We make this precise.

Contributions.

- We formalize PEAM (Section 3): a group-additive classifier with flexible, calibrated per-group experts and exact additive log-odds reason codes, and we state the inductive bias it encodes.
- We give a specification analysis (Section 4) identifying the regime in which PEAM is well-specified and GA2M is not, and the regime (cross-group interactions) in which PEAM is misspecified.
- We verify the analysis with a controlled synthetic study (Section 5) that sweeps interaction order and sample size: at order ≥ 3 PEAM is the single best model, beating EBM and strong, tuned black boxes from n=2k to n=80k.
- We map where the regime occurs in public data (Section 6). On MADELON (Guyon, 2003) PEAM is the best interpretable model and beats full-feature black boxes; on HIGGS (Baldi et al., 2014) its high-order-but-delocalized signal makes PEAM fail, as predicted; on saturated / low-order data (satimage, gas-drift) standard methods win.
- We show that the adequacy of a grouping is structure-dependent (Section 7): correlation grouping succeeds exactly when the interacting features are mutually correlated and fails otherwise, and univariate mutual information fails on jointly-informative features.

All code, synthetic generators, and scripts to reproduce every number in this paper are released with the project.

2 Related work

Additive and glass-box models. GAMs (Hastie & Tibshirani, 1990) model the log-odds as a sum of univariate shape functions. GA2M (Lou et al., 2013) adds a controlled number of pairwise terms, and the Explainable Boosting Machine (Caruana et al., 2015; Nori et al., 2019) is its widely-used gradient-boosted realization and our primary interpretable baseline. Neural additive models (Agarwal et al., 2021) replace the shape functions with neural networks but retain additivity in main (and optionally pairwise) effects. All of these are, by design, limited to low-order interactions. PEAM departs by being additive across *groups* of features while placing an unconstrained learner *within* each group, trading per-feature shape plots for group-level reason codes in exchange for arbitrary within-group interaction order.

Mixtures of experts. PEAM's experts echo the mixture-of-experts lineage (Jacobs et al., 1991), but classical MoE uses a learned *gate* to select experts per input, which obscures attribution. PEAM instead assigns experts to fixed feature groups and combines them additively, so that the contribution of each expert is an exact, globally-consistent reason code. Recent work on interpretable MoE (Ismail et al., 2023) shares the goal of transparent expert combination; our combiner is a transparent linear model in log-odds space.

Log-odds attributions and faithfulness. Working in log-odds (the natural additive scale for a logistic model) connects PEAM to recent arguments for additive evidence in interpretable modelling (Dervovic et al., 2022). Unlike SHAP (Lundberg & Lee, 2017) and LIME (Ribeiro et al., 2016), whose attributions approximate a fixed black box and can disagree with it, PEAM's reason codes are *definitional*: they are the terms of the model's own decision function and sum to it exactly.

3 The Pattern-Expert Additive Model

Setup. Let $x \in \mathbb{R}^d$ with binary label $y \in \{0, 1\}$ and base rate $\pi = \Pr(y{=}1)$. Fix a partition of the feature indices into K groups $G_1, \ldots, G_K$ (disjoint, covering all features). Write x_{G_k} for the sub-vector of features in group k, and let σ be the logistic function and $\text{logit} = \sigma^{-1}$.

Experts. For each group we fit a calibrated expert $e_k : \mathbb{R}^{|G_k|} \to (0, 1)$ estimating $\Pr(y{=}1 \mid x_{G_k})$. In this paper each expert is a histogram gradient-boosted tree trained on the training split and calibrated by Platt scaling (Platt, 1999) on a held-out validation split, so that it reports a probability without leakage from the calibration data into the base fit. Any model class can serve as an expert; the only requirement is a calibrated probability.

Evidence and combiner. Define the group evidence

$$z_k(x) \;=\; \text{logit}\, e_k(x_{G_k}) \;-\; \text{logit}\, \pi, \tag{2}$$

i.e. how much group k moves the log-odds away from the prior. PEAM combines evidence linearly,

$$\text{logit}\; p(x) \;=\; b_0 + \sum_{k=1}^{K} \beta_k\, z_k(x), \tag{3}$$

with (β_k, b_0) fit by logistic regression on the validation split (the *logistic* combiner). A parameter-free *naive* variant sets $\beta_k{=}1$ and $b_0 = \text{logit}\, \pi$, recovering a naive-Bayes-style addition of evidence.

Reason codes. The per-group reason code is

$$r_k(x) \;=\; \beta_k\, z_k(x), \qquad \text{logit}\; p(x) = b_0 + \sum_{k=1}^{K} r_k(x). \tag{4}$$

This decomposition is *exact by construction*: the reason codes are the summands of the model's own decision function, not an approximation of it. A positive r_k is signed evidence from group k for the positive class; the global importance of a group is $\mathbb{E}\,|r_k|$. Groups can be switched off (drop r_k) for deployment in settings that lack a feature group, giving a form of modular portability.

Properties. (i) *Exact additivity* of reason codes, as above. (ii) *Arbitrary within-group interaction order*: each e_k may model any interaction among the features of its group. (iii) *Across-group additivity*: Eq. (3) assumes no interaction between groups — this is PEAM's inductive bias and, as we show, both its strength and its failure mode. (iv) *Calibration*: per-expert calibration plus a logistic combiner yields well-calibrated probabilities (Section 8).

Algorithm 1 PEAM training (logistic combiner)

Require: train (X^{tr}, y^{tr}), validation (X^{val}, y^{val}), groups $\{G_k\}$
1: $\pi \leftarrow \text{mean}(y^{tr})$
2: **for** $k = 1, \ldots, K$ **do**
3: fit base learner on $(X^{tr}_{G_k}, y^{tr})$
4: calibrate it on $(X^{val}_{G_k}, y^{val})$ to obtain expert e_k
5: **end for**
6: form evidence matrix Z^{val} with $Z^{val}_{ik} = \text{logit}\, e_k(x^{val}_{i,G_k}) - \text{logit}\, \pi$
7: fit logistic regression of y^{val} on Z^{val} to obtain (β, b_0)
8: **return** $\{e_k\}, \beta, b_0, \pi$

4 When does PEAM win?

Suppose the true conditional log-odds decomposes as

$$\text{logit}\, \Pr(y{=}1 \mid x) \;=\; c_0 + \sum_{k=1}^{K} f_k(x_{G_k}), \tag{5}$$

i.e. it is additive *across* the groups $\{G_k\}$ but each f_k is an arbitrary function of its group. Then:

- **PEAM is well-specified.** With a sufficiently flexible expert per group, e_k can match f_k up to calibration, and the additive combiner in Eq. (3) matches the across-group structure. PEAM can represent Eq. (5) regardless of the *order* of each f_k.
- **GA2M / EBM is misspecified** whenever some f_k contains a genuine interaction of order ≥ 3: its hypothesis class (main effects plus pairwise terms) cannot express a third-order term, so it incurs irreducible approximation error no matter how much data it sees.
- **Linear models are misspecified** whenever any f_k is nonlinear.

Conversely, if the truth contains interactions that *cross* groups — a term depending jointly on features from G_j and G_k — then PEAM's across-group additivity (iii) is violated and PEAM is itself misspecified, while a black box or even GA2M (through a cross-group pairwise term) may do better. Hence the operative condition is not high interaction order alone but *high order* ***and*** *locality*: the interactions must live inside the groups. We test both halves of this prediction below.

A second mechanism matters in practice. When many features are pure noise and the informative features fall in one group, the logistic combiner drives the noise groups' weights $\beta_k \to 0$, so PEAM performs *implicit structured feature selection*. This lets PEAM beat a monolithic black box that must find the informative subset among all features — a point borne out on MADELON.

5 Controlled study: sweeping interaction order

Generator. We generate data in which the per-group signal is a *pure* order-r interaction. For K groups of m features each, draw all features i.i.d. $\mathcal{N}(0,1)$ (independent within and across groups) and set

$$z_k = \text{standardize}\left(\textstyle\prod_{j=1}^{r} x_{k,j}\right), \qquad \text{logit}\, p = b + \textstyle\sum_{k=1}^{K} w\, z_k, \qquad y \sim \text{Bernoulli}(\sigma(\text{logit}\, p)), \tag{6}$$

with b solved to fix the positive rate (we use 10%). A product of r independent mean-zero variables has zero correlation with any single feature and with any lower-order product of its own factors, so the signal requires modelling the *full* order-r interaction; the log-odds is additive across groups by construction, so Eq. (5) holds exactly. We add pure-noise features to make grouping non-trivial.

Table 1: Synthetic interaction-order sweep (K=4 groups, 10% positives, mean over 3 seeds). PEAM uses the true grouping. `gbm` is the non-interpretable black-box reference; all others are interpretable.

	ROC-AUC			
interaction order	logreg	EBM	**PEAM**	gbm (black box)
1 (linear)	0.936	0.930	0.868	0.918
2 (pairwise)	0.515	0.908	0.856	0.890
3 (localized high-order)	0.485	0.668	**0.795**	0.731
4 (localized high-order)	0.522	0.679	**0.685**	0.637
	PR-AUC			
1	0.686	0.667	0.497	0.619
2	0.121	0.618	0.472	0.555
3	0.104	0.183	**0.418**	0.312
4	0.128	0.179	**0.254**	0.166

Table 2: Order-3 robustness (ROC-AUC, mean over 3 seeds). PEAM uses the true grouping; the black boxes are strong and tuned (HistGBM: 2000 trees, early stopping; XGBoost: 1500 trees, depth 6).

n	**PEAM**	EBM	HistGBM (strong)	XGBoost
2,000	**0.651**	0.636	0.603	0.622
8,000	**0.795**	0.668	0.733	0.717
30,000	**0.884**	0.725	0.856	0.845
80,000	**0.905**	0.746	0.889	0.886

Setup. K=4 groups of 4 features, 6 noise features, 10% positives, stratified 60/20/20 train/validation/test, three seeds. We compare logistic regression, EBM (the interpretable baseline), PEAM with the true grouping, and a black-box histogram GBM. Experts and the GBM use comparable capacity.

Result (Table 1). There is a clean crossover. At order 1 the truth is linear and logistic regression and EBM win; PEAM's flexible experts are unnecessary overhead. At order 2 the truth is exactly GA2M's design point and EBM wins. At order ≥ 3 EBM saturates — it cannot represent the interaction — while PEAM with the true grouping becomes the *single best model, above the black box*, on both ROC-AUC and PR-AUC, because its inductive bias (across-group additivity, free within-group order) matches the data exactly whereas the black box must discover the group structure from limited data.

Robustness. Is PEAM's edge over the black box a tuning or small-sample artifact? Table 2 fixes order 3 and gives the black boxes every advantage: a strong early-stopped histogram GBM and XGBoost, across sample sizes. PEAM leads at every n. The black box closes the gap as n grows (it eventually learns the block structure) but never overtakes the model that is told it; EBM never catches up because order-3 terms lie outside its hypothesis class.

6 Real-data study: where the regime occurs

We now ask whether the regime arises in public data. Table 3 summarizes four datasets chosen to span the possibilities; we describe the grouping used for each.

MADELON: a localized high-order win. MADELON (Guyon, 2003) has 500 features, of which only 5 are informative — placed on the vertices of a five-dimensional hypercube, so the class is defined by a five-way interaction — plus 15 redundant features (random linear combinations of the 5) and 480 noise probes. This is precisely a localized high-order interaction embedded in noise. PEAM with correlation grouping (Section 7)

Table 3: Real-data hunt (ROC-AUC, mean over 3 seeds). `peam_*` are interpretable; `xgb_*` is the black-box reference. Groupings are domain-defined except where noted.

dataset	model	ROC-AUC
satimage (9 px × 4 bands)	EBM	**0.961**
	PEAM (semantic groups)	0.948
	PEAM (correlation groups)	0.941
gas-drift (16 sensors × 8)	logistic regression	0.999
	EBM	1.000
	PEAM (semantic groups)	0.999
HIGGS (per-object kinematics)	EBM (low-level only)	**0.674**
	PEAM (per-object groups)	0.608
	XGBoost (low-level only)	0.680
	EBM (with cross-object masses)	0.796
MADELON (5 informative + noise)	EBM (full)	0.800
	XGBoost (full)	0.857
	PEAM (correlation groups, full)	**0.903**
	EBM (shared feature selection)	0.885
	XGBoost (shared feature selection)	0.927
	PEAM (shared feature selection)	**0.912**

attains ROC-AUC 0.903, beating EBM (0.800) and full-feature XGBoost (0.857); it is the best interpretable model by a wide margin and the best model overall *except* for a black box equipped with a separate feature-selection stage. To test fairness we gave every model the same ExtraTrees-importance feature selection (≈ 14 features): even then PEAM (0.912) beats EBM (0.885), because selection removes noise but does not grant EBM the fifth-order term it structurally lacks. Only the select-then-boost black-box pipeline (0.927) edges PEAM — at the cost of interpretability and an extra stage. The win is stable: over five seeds PEAM averages 0.895 (range $[0.863, 0.936]$) versus EBM 0.795, Random Forest 0.768, and ExtraTrees 0.747.

HIGGS: high-order but delocalized — the contrast. HIGGS (Baldi et al., 2014) is the canonical example of a dataset whose high-order structure defeats additive models. Its 21 low-level features are per-object kinematics (lepton, missing energy, four jets); its 7 high-level features are *invariant masses* that combine *multiple* objects. Grouping by object and applying PEAM to the low-level features gives ROC-AUC 0.608, *below* EBM's 0.674: the discriminative interactions cross object boundaries, violating PEAM's across-group additivity. The point is sharpened by the last row: adding the seven cross-object mass features lifts EBM from 0.674 to 0.796. HIGGS has high-order signal, but it is *delocalized*, so PEAM fails exactly as the analysis in Section 4 predicts. Locality, not interaction order alone, is what PEAM requires.

satimage and gas-drift: not in the regime. On satimage (remote sensing; a 3×3 pixel neighbourhood over 4 spectral bands) EBM already reaches 0.961 and PEAM with the genuine pixel/band grouping is slightly behind: the signal is well captured by additive and pairwise terms. On gas-drift (16 chemical sensors ×8 descriptors) every model, including logistic regression, exceeds 0.99 — the task is near-separable, leaving no interaction headroom. These are the common case for engineered tabular features, and they are *not* where PEAM helps.

7 Which grouping works?

PEAM requires a grouping that places the interacting features together. We find that the adequacy of a grouping is *structure-dependent*, and that no single automatic method is universal.

- **Correlation grouping** (hierarchical clustering on |Spearman|) recovers the interacting set exactly when those features are mutually correlated. On MADELON the 15 redundant features are linear combinations of the 5 informative ones, so the useful set forms a correlated cluster and correlation grouping isolates it from the 480 noise probes — which is why PEAM wins there. In the synthetic generator, by contrast, the interacting features are independent products with *zero* pairwise correlation, so correlation grouping cannot find them and PEAM with correlation (or random) grouping collapses to near the additive baseline.
- **Univariate mutual information** fails on jointly-informative features: a pure high-order interaction has (near) zero marginal mutual information with the label, so ranking features by univariate MI — the classic "MADELON trap" — selects the wrong features and PEAM gains nothing.

The practical consequence is that PEAM is most readily deployed when a grouping is given by domain knowledge, and that *interaction-aware* group discovery (rather than correlation or marginal association) is the right open problem for making PEAM automatic.

8 Interpretability and calibration

PEAM's reason codes are exact (Section 3); we also check that they are *faithful* and that the model is *calibrated*. On a suite of imbalanced public datasets, zeroing the evidence of the single most important group at test time drops ROC-AUC by ≈ 0.14 on average, confirming that the reason codes identify genuine drivers rather than redundant signal. PEAM's expected calibration error (≈ 0.014) is essentially tied with EBM's, the best among the models we test, as a consequence of per-expert Platt calibration followed by the logistic combiner (Platt, 1999; Guo et al., 2017). Interpretability is at the *group* level: PEAM reports which group drove a decision and by how much, exactly, but — unlike EBM's per-feature shape functions — it does not resolve attribution *within* a group, whose expert is a local black box. This is the representational price of capturing within-group high-order interactions.

9 Limitations

We state the boundaries of the contribution plainly. (i) PEAM requires a grouping that captures the interacting features; obtaining one automatically is unsolved (Section 7). (ii) PEAM is dominated on linear, pairwise, saturated, or cross-group-interaction data — a large fraction of real tabular problems. (iii) A black box combined with a separate feature-selection stage can exceed PEAM even in its favourable regime (MADELON: 0.927 vs 0.912), trading interpretability for a small accuracy gain. (iv) Interpretability is group-level, not per-feature. PEAM is therefore best understood as a targeted instrument for a specific, identifiable regime, not a replacement for GA2M or gradient boosting in general.

10 Conclusion

We introduced PEAM, a group-additive classifier with flexible, calibrated per-group experts and exact additive log-odds reason codes, and we characterized — honestly, including where it loses — a previously uncharted point on the accuracy–interpretability frontier. When the predictive signal consists of high-order interactions localized within a known grouping, PEAM is the best interpretable model and can beat strong black boxes; we demonstrated this in a controlled order-sweep and on the MADELON benchmark, and we showed via HIGGS that locality, not interaction order alone, is the operative condition. The natural next step is interaction-aware group discovery, which would turn PEAM from a model that *exploits* a known grouping into one that *finds* the regime on its own.

Reproducibility. The PEAM implementation, the synthetic generator, and scripts that regenerate every table (`run_interaction_locality.py`, `run_real_data_hunt.py`) are released as open source at `https://github.com/rupeshbharambe24/peam-experts`. All real datasets are public and fetched without credentials. Experiments run on CPU.

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